

Given,

```
int fib(int n){  
    if (n==0)  
        return 0;  
    else if (n==1)  
        return 1;  
    else  
        return fib(n-1) + fib(n-2);  
}
```

Let,

$n = \$t_0$

address

~~pointer~~

~~return~~ return value = \$s8

return value = \$t4, pointer = \$p

Mips code:

fun:

addi \$p, \$p, -8	1	2
sw \$s8, 4(\$p)	3	3
sw \$t0, 0(\$p)	4	4
beq \$t0, \$0, r1	5	5
li \$t1, 1	6	6
beq \$t0, \$t1, r2	7	7
sub \$t0, \$t0, \$t1;	8	8
Jal fun;	9	9
add move \$t3, \$t4, \$0	10	10
lw \$t0, 0(\$p)	11	11
li \$t1, 2	12	12
sub \$t0, \$t0, \$t1	13	13

jal fun 14

add \$t4, \$t4, \$t3 15

end 16

r1: 17

li \$t4, 0 18

end 19

r2: 20

li \$t4, 2 21

end 22

end: 23

lw \$r8, 0(\$p) 24

addi \$p, \$p, 8 25

jr \$r8; 26

—X—

total time = 26